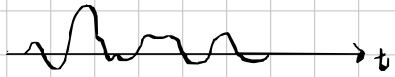
Mode

Activator: an optical mode is a distinct pattern of electromagnetic oscillation, characterized by its spatial, temporal, frequency, and polarization. Modes are the fundamental building blocks of an electromagnetic field. Just as a complex sound can be decomposed into individual frequencies, an arbitrary electromagnetic field can be expressed as a superposition of modes.

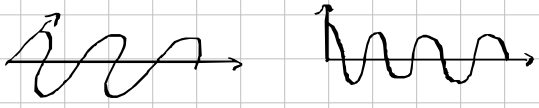
Consider an arbitrary field



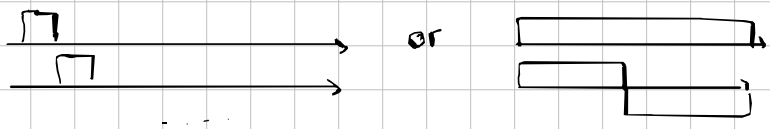
We can decompose this as spatial modes, frequency, ...



We can also decompose into polarizations



We can also define modes such as



Anything can be considered as a mode as long as the quantity is preserved throughout the lifetime and they satisfy

- Normalization

$$\int_A \int_0^T \phi_v(\vec{r},t)\,\phi_v^*(\vec{r},t)\,d\vec{r}\,dt = 1$$

- Orthogonalization

$$\int_A \int_0^T \phi_v(\vec{r},t)\,\psi_\mu^*(\vec{r},t)\,d\vec{r}\,dt = 0$$

For a bandwidth W, there can be TW or

Coherent State

A coherent state is only realistic for a finite window. A coherent state is used to define the frequency mode of an amplitude. Consider

$$E(t,\vec{r}) = \mathcal{E}(t,\vec{r})\, \alpha\, e^{i(\omega t + \varphi)}$$